Graph Neural Networks in Computer Vision

Hands-on Workshop on Deep Learning for Computer Vision
MG University

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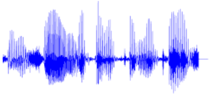
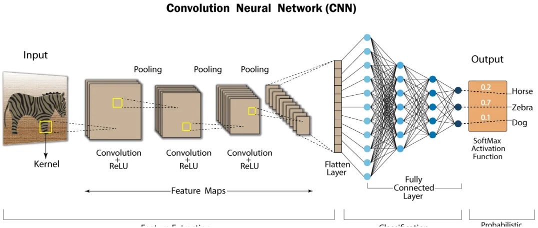
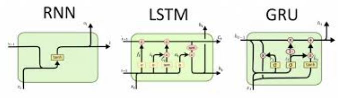
Overview

1. Introduction
2. Why are graphs useful?
3. Why is it difficult to define convolution on graphs?
4. What makes a neural network a graph neural network?

Traditional Neural Networks



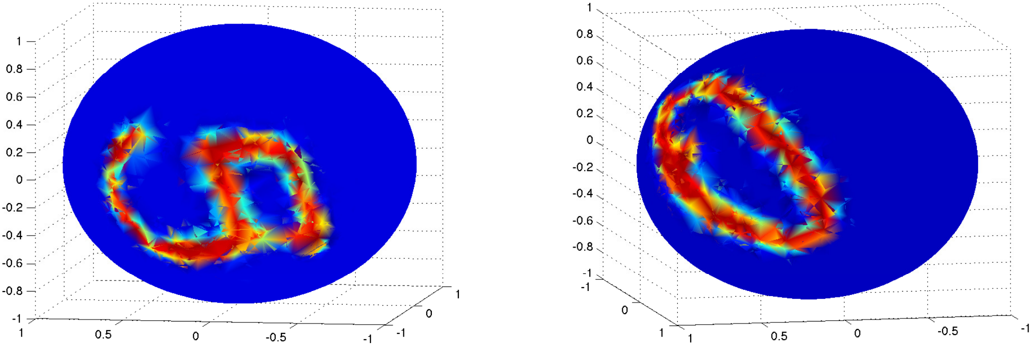




Deep neural networks are designed for sequences and grids exploiting

- Translation equivariance (weight sharing)
- Hierarchical compositionality

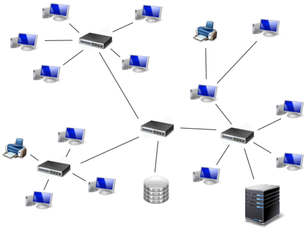
Real-world data are rarely on grids!



Examples of some MNIST digits on the sphere.[1]

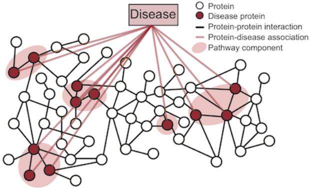
[1] J Bruna et al. (2014). "Spectral networks and deep locally connected networks on graphs, 2nd Int". In: *Conf. on Learning Representations, ICLR 2014—Conference Track Proceedings*.

Social Networks

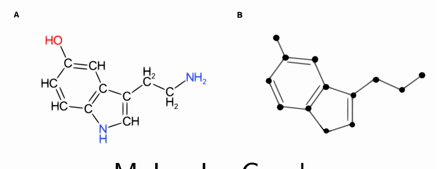


Computer Networks

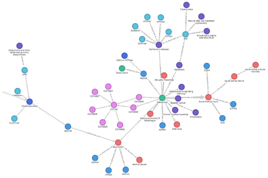
Transportation Networks



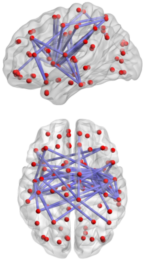
Disease Pathways



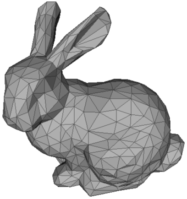
Molecular Graphs



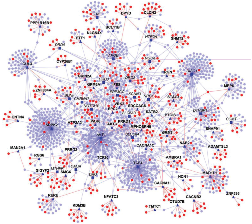
Knowledge Graphs



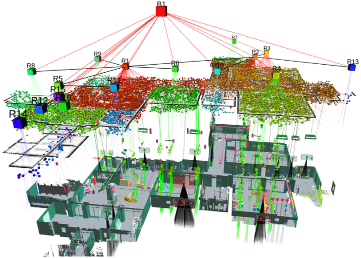
Brain Networks



3D Shapes



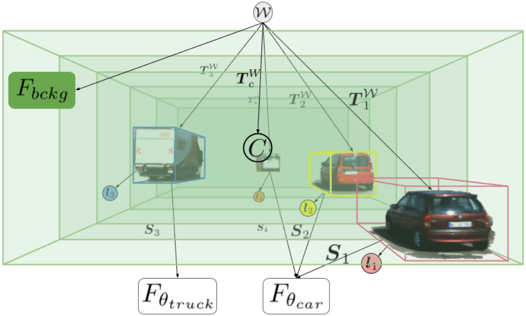
Protein Interaction Networks



3D Scene Graphs

Rethinking Convolutions: The Graph Way

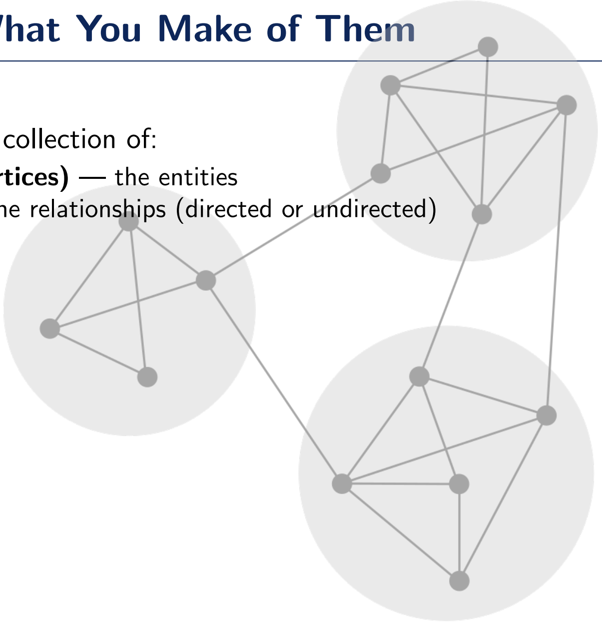
- Why are graphs useful?
- Why is it difficult to define convolution on graphs?
- What makes a neural network a graph neural network?



Ost et al. 2021

Graphs Are What You Make of Them

- A graph G is a collection of:
  - **Nodes (vertices)** — the entities
  - **Edges** — the relationships (directed or undirected)



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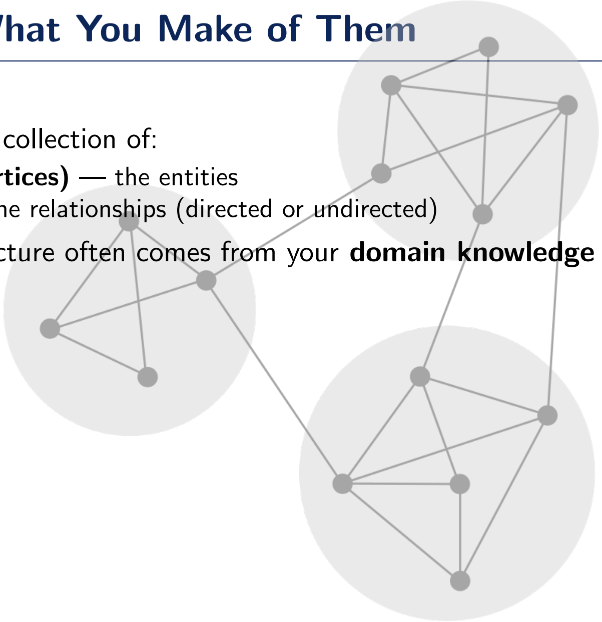

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- You decide:
  - Atoms $\leftrightarrow$ Molecules
  - Users $\leftrightarrow$ Social Networks
  - Cities $\leftrightarrow$ Transportation Systems
  - Players $\leftrightarrow$ Team Sports
  - Neurons $\leftrightarrow$ Brain Networks
  - Objects $\leftrightarrow$ Physical Systems
  - Pixels, Bounding Boxes $\leftrightarrow$ Images

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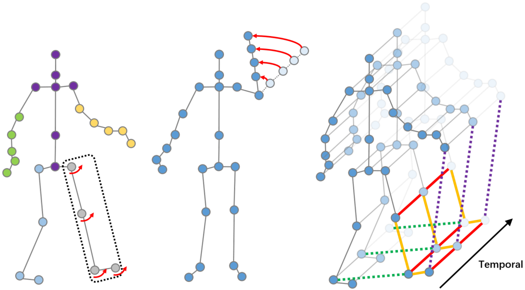
  - Pixels, Bounding Boxes $\leftrightarrow$ Images
- *"In practice, you define what the nodes and edges represent."*

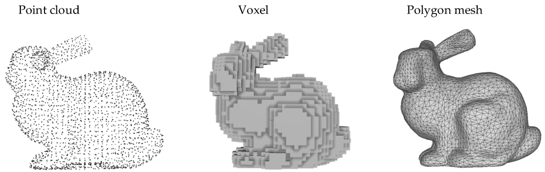
Graphs as Alternative Data Representations

- Provide alternative and flexible data representations.
- Many CV/ML tasks can benefit from viewing data as graphs rather than arrays or tensors.
- Graphs offer flexibility in capturing spatial and semantic relationships.
  - Superpixel graphs in images (Liang et al. 2016)
  - Human pose as a skeleton graph (Yan, Xiong, and Lin 2018)
  - Videos as graphs of moving objects (Wang and Gupta 2018)
  - Faces via facial landmark graphs (Antonakos, Alabort-i-Medina, and Zafeiriou 2015)



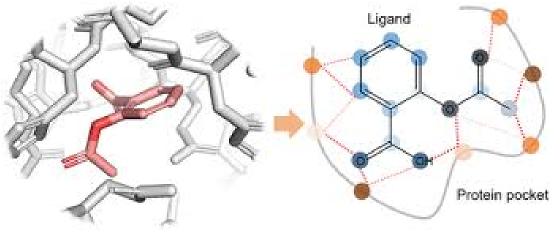
Graphs are Natural for Many Learning Tasks

- Many problems are naturally graph-structured:
  - Molecule and social network classification (Knyazev et al. 2018)
  - 3D mesh processing (Fey et al. 2018)
  - Dynamic systems modeling (T. Kipf et al. 2018)
  - Visual scene graphs and reasoning (Narasimhan, Lazebnik, and Schwing 2018)
  - Program synthesis and RL tasks (Bapst et al. 2019)
- GNNs empower learning from these rich relational structures.

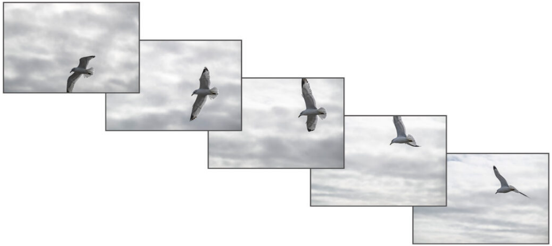


A Gateway to Solving Complex Real-World Problems

- Enables tackling previously unsolvable real-world problems:
  - Drug discovery (Veselkov et al. 2019)
  - Human brain connectivity (Diez and Sepulcre 2018)
  - Materials discovery for energy and environmental challenges (Xie et al. 2019)

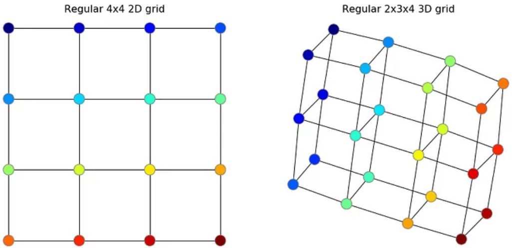


Why is convolution useful?



- CNNs exploit:
  - Shift invariance
  - Locality
  - Compositionality (or hierarchy)
  - Convolutional layers are parametric

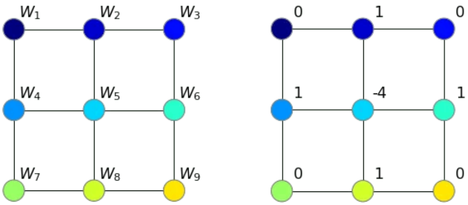
Convolution on Images in Terms of Graphs



Examples of regular 2D and 3D grids.

- Let's apply conv filters on this

Convolution on Images in Terms of Graphs

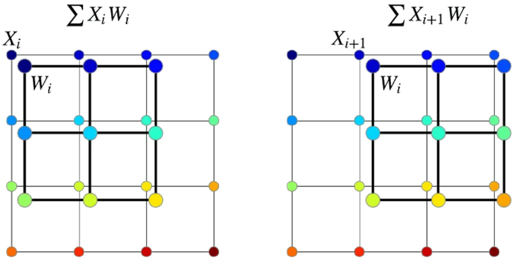


3×3 filter on a regular 2D grid with arbitrary weights w and an edge detector.

- We slide over all possible locations.
- We compute the dot product:

$$W : X_1 W_1 + X_2 W_2 + ... + X_9 W_9$$

Convolution on Images in Terms of Graphs

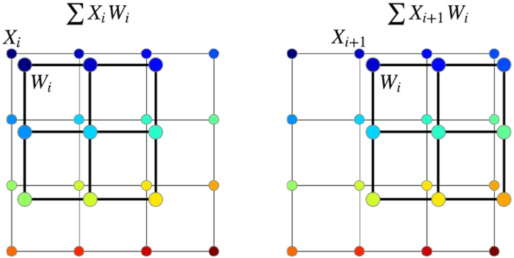


2 steps of 2D convolution on a regular grid.

- The dot product is an aggregator operator.
- It summarizes a 3×3 matrix to a single value.
- But is not permutation invariant. i.e.

$$X_1 W_1 + X_2 W@ \neq X_1 W_2 + X_2 W_1$$

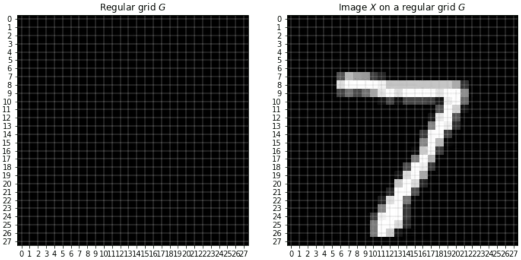
Convolution on Images in Terms of Graphs



2 steps of 2D convolution on a regular grid.

- In a regular grid, we always can match a node of the filter with a node of the grid.
- Let's see this in action on MNIST.

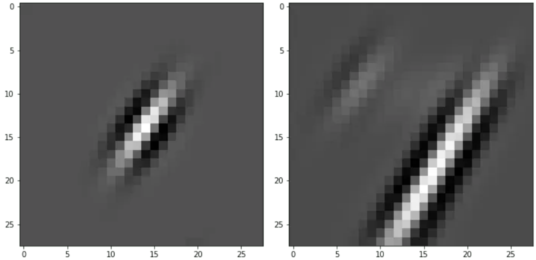
Convolution on Images in Terms of Graphs



Regular 28×28 grid and an image on that grid.

- Define a filter with arbitrary parameters and perform convolution.

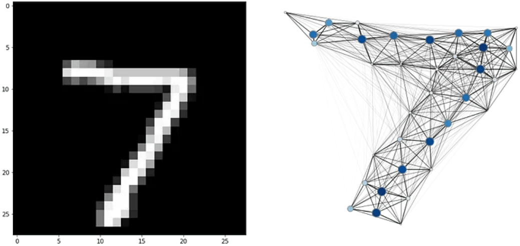
Convolution on Images in Terms of Graphs



A 28×28 filter and the result of 2D convolution of this filter with the image of digit 7.

- Can we generalize convolution to graphs?

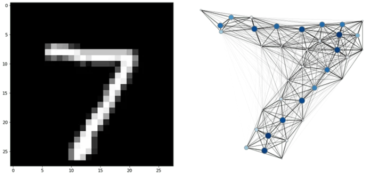
Convolution on Images in Terms of Graphs



An image from the MNIST dataset and its graph representation.

- Nodes in a graph form an unordered set.
- Any permutation of the node set does not change the graph.
- Therefore, the aggregator must be **permutation-invariant**.
- In graphs, there's no consistent node order — unless you learn or enforce one.

Convolution on Images in Terms of Graphs



An image from the MNIST dataset and its graph representation.

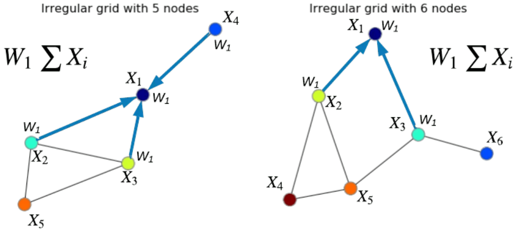
- Solution: Use permutation-invariant aggregators.
  - [2]: Mean aggregation
  - [3]: Sum aggregation
  - [4]: GraphSAGE

[2] Thomas N Kipf and Max Welling (2016). "Semi-supervised classification with graph convolutional networks". In: *arXiv preprint arXiv:1609.02907*.

[3] Keyulu Xu et al. (2018). "How powerful are graph neural networks?" In: *arXiv preprint arXiv:1810.00826*.

[4] Will Hamilton, Zhitao Ying, and Jure Leskovec (2017). "Inductive representation learning on large graphs". In: *Advances in neural information processing systems* 30.

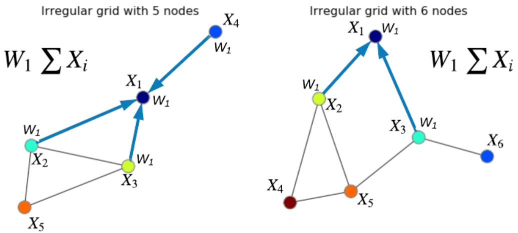
Convolution on Images in Terms of Graphs



Convolution on graphs of node features X with filter W centered at node 1

- Node 1: $h_1 = (x_1 + x_2 + x_3 + x_4)W$
- Node 2: $h_2 = (x_1 + x_2 + x_3 + x_5)W$
- Continue similarly for other nodes...

Convolution on Images in Terms of Graphs



Convolution on graphs of node features X with filter W centered at node 1

- Use the **same weight matrix** W for all nodes.
- Produces **new node features** enriched with neighbor information.
- The graph structure remains unchanged, only the node features are updated.
- **Stack multiple layers** to let information flow deeper into the graph.

What makes a neural network a graph neural network?

- In a vanilla neural network, the output of layer l is:

$$X^{(l+1)} = X^{(l)} W^{(l)}$$

- But this operation ignores graph structure — it's just applied row-wise.
- How do we transform this to a graph neural network?

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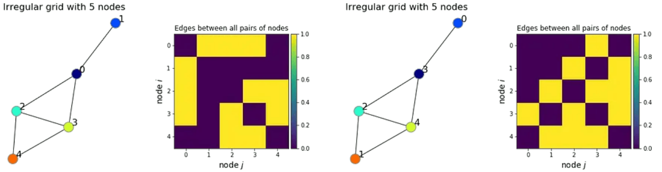
- But this operation ignores graph structure — it's just applied row-wise.
- How do we transform this to a graph neural network?
- The core idea behind GNNs is aggregation over **neighbors** specified by *you*.

What makes a neural network a graph neural network?



The order of nodes we defined in both cases is random, while the graph is still the same.

- Adjacency matrix represents edges in matrix form.
- Each row/column in A corresponds to a node.
- This encodes “who is connected to whom”.
- It is the prior/inductive bias we impose on our model.

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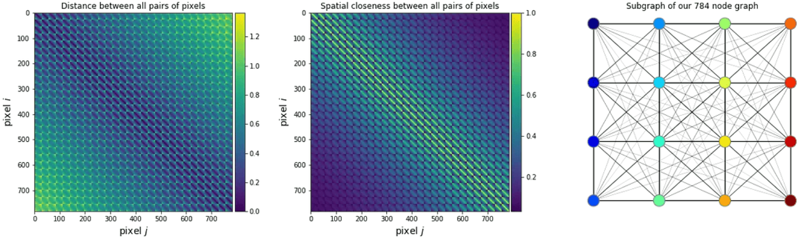
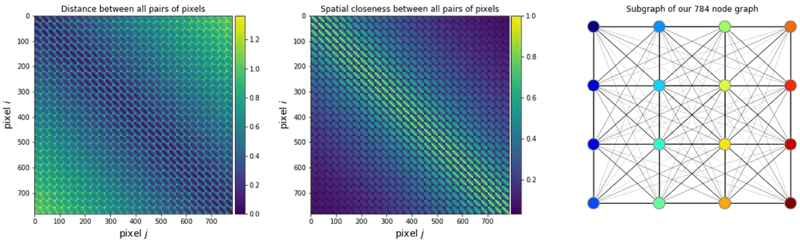


Adjacency matrix (NxN) in the form of distances and closeness between all pairs of nodes. A subgraph with 16 neighboring pixels corresponding to the adjacency matrix in the middle.

- Instead of having just features X we have a matrix A.
- But there is no canonical order of nodes that will be consistent across all other graphs in the dataset.

What makes a neural network a graph neural network?



Adjacency matrix (NxN) in the form of distances and closeness between all pairs of nodes. A subgraph with 16 neighboring pixels corresponding to the adjacency matrix in the middle.

- Regular MLPs assume the order of input features matters (like pixels in an image).
- In graphs, node order is arbitrary, so that assumption breaks.
- Using the adjacency matrix A helps us do learning in a way that doesn't care about node order — and that's what Graph Neural Networks are built for.

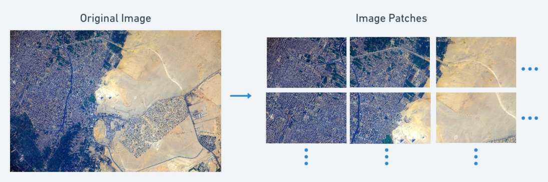
Graph Neural Networks

- We want to update the features of each node by combining info from its neighbors — this is **message passing**.
- Multiply the adjacency matrix with the feature matrix

$$X^{(l+1)} = \mathcal{A}X^{(l)}W^{(l)}$$

  where $\mathcal{A} = \frac{A}{\sum_i A_i}$
- This is the core of a Graph Convolutional Network (GCN).

Other Visual Node representations

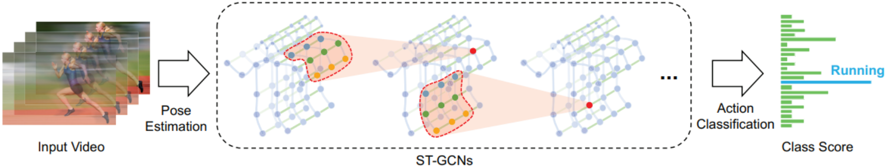


- Split an image into fixed-size patches and view as nodes.
- Parse images into semantic graphs of objects and relationships — enabling structured understanding for downstream tasks.

[5]

[5] Yikang Li et al. (2019). "Pastegan: A semi-parametric method to generate image from scene graph". In: *Advances in Neural Information Processing Systems* 32.

Other Visual Node representations

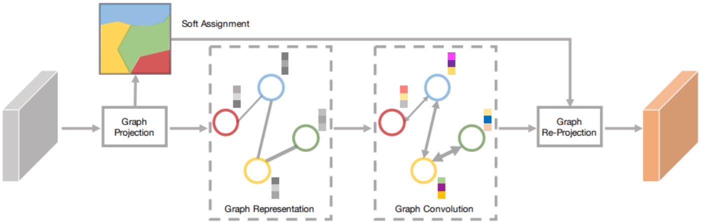


- Human joints linked by skeletons naturally form a graph and learn human action patterns.

[6]

[6] Sijie Yan, Yuanjun Xiong, and Dahua Lin (2018). "Spatial temporal graph convolutional networks for skeleton-based action recognition". In: *Proceedings of the AAAI conference on artificial intelligence*. Vol. 32. 1.

Other Visual Node representations

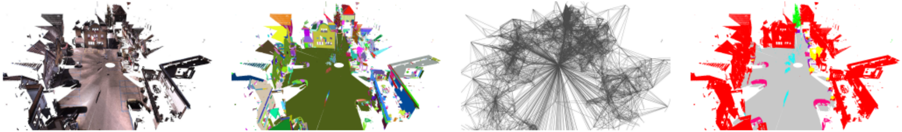


- Assign pixels with similar features to the same vertex grouping pixels into coherent regions.

7

[7] Yin Li and Abhinav Gupta (2018). "Beyond grids: Learning graph representations for visual recognition". In: *Advances in neural information processing systems* 31.

Other Visual Node representations



(a) RGB point cloud (b) Geometric partition (c) Superpoint graph (d) Semantic segmentation

- Aggregate k-nearest neighbors into superpoints (nodes), then use GCNs to model their relationships and understand the topology of the surrounding environment.

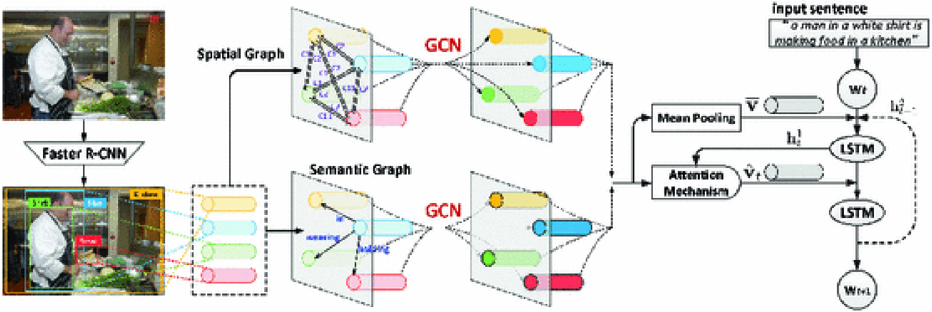
[8]

[8] Loic Landrieu and Martin Simonovsky (2018). "Large-scale point cloud semantic segmentation with superpoint graphs". In: *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 4558–4567.

Visual Edge Representation in GNNs

- **Edges capture spatial and temporal relations** between visual elements (e.g., objects, regions, patches).
- **Spatial Edges:**
  - Fixed using scene graphs or human skeletons (static).
  - Learned dynamically via self-attention or fully-connected graphs.
- **Temporal Edges:**
  - Link nodes across frames using k-NN or semantic similarity.
  - Dynamic methods like random walks (Markov chains) model evolving node relations.
- GNNs (spectral/spatial) leverage these edges to understand images and video over space and time.

Visual Caption, QA



- After detecting image regions a semantic/spatial graph is built where nodes are regions and edges capture relationships.
- The relation-aware features are passed through attention-based LSTM decoders to generate captions.

9

[9] Ting Yao et al. (2018). "Exploring visual relationship for image captioning". In: *Proceedings of the European conference on computer vision (ECCV)*, pp. 684–699.

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